Augmenting Live Broadcast Sports with 3D Tracking Information

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Techniques that enhance television sports broadcasts use algorithms for tracking objects, such as the ball, to create informative, graphical visualizations embedded into the broadcast image.

In recent years, Sportvision has developed technologies that provide an augmented experience for viewers of network and cable broadcast sporting events. The final televised video contains enhancements that assist the viewer in understanding the game and the strategies of players and teams (see Figure 1). Some effects enable advertisements and logos to be integrated into the on-air video in subtle ways that don't distract the viewer. Example visuals include graphical aids that help the audience easily locate rapidly moving objects or players across the ice in hockey or the ball in football and baseball; information such as the speed or predicted trajectories of interesting objects such as a golf ball, golf club, or a baseball or bat; and highlights of particular locations within the arena, such as in soccer, marking the offside line, which moves dynamically depending on the gameplay.

A cornerstone of our particular approach is a careful calibration of broadcast cameras (those that gather the images that are sent to the viewing public) so that we precisely know the image's spatial coverage.[1,2] A separate but equally important aspect of the approach is the sensor apparatus and processing that provides a means for locating and tracking important objects in the sporting event—for instance, the locations of the puck and players in a hockey game.[3]

The raw video images and their augmentations (the graphics) are merged using a technique similar to the chroma-key process.[4] However, our compositing process is more capable than standard chroma-keying because our approach allows translucent (or opaque) graphical elements to be inserted between foreground and background objects (the augmented objects are considered background objects). A novelty is that our approach allows for specifying multiple ranges of colors to be keyed, and some specifically not keyed. This enables the addition of graphics such as corporate logos or other graphical details (for example, in soccer allowing a virtual off-sides line on a field in such a way that the graphics appear to have been painted on the grass).

Our work is driven by the demands of network television broadcasting and is accordingly focused on implementations that run in real time and are economically feasible and practical. Nevertheless, many of these ideas can be applied in other arenas such as augmented reality.[5] We will describe several applications of the approach; some incorporate computer vision to track objects while others use nonvision sensors but serve to illustrate important aspects of our general approach.

In this article, we describe the evolution of our tracking system and broadcast effects from 1996 to 2006. Our initial system was successfully deployed during the 1996 National Hockey League (NHL) all-star game, enhancing televised live hockey with a visual marker (a bluish glow) over the puck in the broadcast image.[1,3,6] Since that time, we've applied our technologies to numerous Fox Sports, ABC, and ESPN live broadcasts including National Football League (NFL) games, Major League Baseball (MLB) games,[7,8] soccer, skating, Olympic sports, and NASCAR racing events,[9]and have been scheduled to appear until at least 2014 by the NFL, MLB, and NASCAR live broadcasts.[10] In this article, we focus on individual tracking technologies and the core instrumentation

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and compositing techniques shared by these various implementations.

Overview: adding virtual elements to a live broadcast

Most of our effects depend on real-time integration of graphical elements together with raw video of the scene to yield a final, compelling broadcast image. Before we discuss methods for integrating dynamically tracked elements into a broadcast image, it's instructive to consider the steps necessary for integrating static graphical elements, such as a logo or a virtual line marker, with raw video.

Figure 2 provides a high-level overview of the flow of video and data elements through our system to insert a virtual graphical element (in this case, a corporate logo) into a live football scene. For the effect to appear compelling, the graphic must be firmly stuck in place in the environment (for example, the ground). Achieving this component of realism proved to be one of the most significant challenges



(a) **(b)**

Figure 1. In the K-Zone system multiple technologies—including tracking, camera instrumentation, and video compositing—come together at once to provide a virtual representation of (a) the strike zone over home plate and occasionally (b) a trail on the ball.

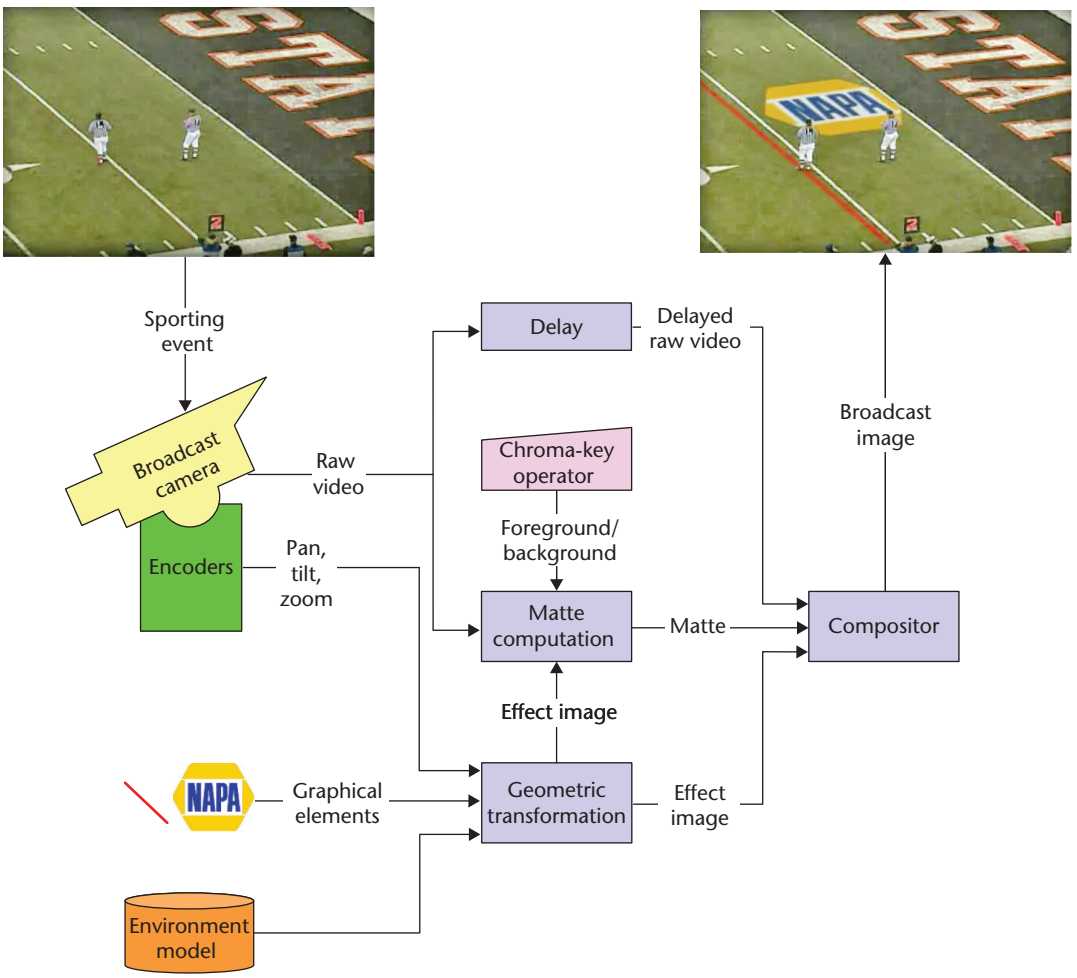


Figure 2. Overview of the video and data flow from the field of play (on left) to the final composited image (on right). Graphical elements are integrated into the live scene using geometric transformations and online matting computations.



Figure 3. Objects that are tracked dynamically can be annotated in the live video. In this example, Indy cars are tracked using GPS technologies; then the positions of different vehicles are indicated with pointers.

for our technology. It requires accurate and precise calibration of broadcast cameras, correct modeling of the environment and fast computer graphics transformations and compositing.

For now, consider the sequence of steps depicted along the bottom of the diagram. A corporate logo is transformed so that it appears in proper perspective in the scene. The transformation step is accomplished using a model of the environment that describes the location of the camera and the relative locations of and orientations of virtual graphical objects to be included. Encoders on the camera are used to calculate the current field of view (FOV) of the camera and perform the appropriate clipping and perspective transformation on the graphical elements.

We must also compute a matte that describes at each pixel whether foreground or background elements are present. The compositor uses the matte to determine an appropriate mixture of the effect image and raw video for the final image.

Adding dynamic objects to the image

Given the infrastructure described previously, it's fairly straightforward to add dynamic elements. The locations of the graphical elements can be determined using any of a number of tracking algorithms as appropriate to the sport.

As an example, we've recently developed a system for tracking the locations of all the competitors in a motor-racing event. Each vehicle is equipped with a GPS receiver, an inertial measurement unit, and other measurement devices. The output of each device is transmitted to a central processor that uses the information to estimate the position of each vehicle.[9]

Given the position information and the pose of the broadcast cameras, we're able to superimpose relevant graphics that indicate the positions of particular vehicles in the scene (see Figure 3).

Camera calibration

Yards lines, puck comet trails, and logos must be added to the video at the proper locations and with the right perspective to look realistic. To determine proper placement, the broadcast camera's FOV is computed from data collected from sensors on the cameras. Others have also addressed this problem and propose to exploit redundant information for auto calibration.[11,12]

To determine the video image's FOV, the tripod heads' broadcast cameras are fitted with optical shaft encoders to report pan and tilt (Figure 4 shows an example tripod head). The encoders output 10,000 true counts per revolution (we obtain 40,000 counts per revolution by reading the phase of the two optical tracks in the encoders). The angular resolution translates to 0.6 inches at 300 feet. Zoom settings are determined by converting an analog voltage emitted by the camera electronics to digital using an analog-to-digital converter.

The broadcast cameras are registered to their environment before the game by pointing them at several reference locations on the field (for example, the 50-yard line in football or home plate in baseball) and annotating their location in the image using an input device such as a mouse click. This registration process enables us to know the precise location and orientation of each broadcast camera.

Video compositor

Many industries such as broadcasting, videography, and filmmaking include techniques for combining images that have been recorded in one environment with those images recorded in another. For example, weather forecasters are depicted in front of an augmented, animated weather map while in reality they're in front of a blue screen; or actors are filmed safely in a studio while they appear to the audience on the South Pole or climbing Mount Everest.

Chroma-keying is a standard technique that replaces the original blue-screened background with that of the special effect—for instance, a



Figure 4. The broadcast camera is equipped with precise shaft encoders that report its orientation and zoom.

more interesting geographical location, giving the illusion of an actor submerged in a more intriguing environment. The resulting image is called the *composite image*. An extension to chroma-keying, typically called *linear keying* (or *digital compositing*) allows the inclusion of transparent objects by proportionally displacing the pixels in one image (the foreground) with corresponding pixels of another image containing the augmented objects (the background). The specific proportion that a pixel is displaced is determined by an alpha signal. The alpha signal determines the opacity of the foreground pixel to the corresponding background pixel. An alpha signal of $\alpha = 0$ indicates that the foreground pixel is transparent, while an alpha signal of $\alpha = 1$ indicates that it's opaque. The values of the alpha signal for all the pixels in an image are given by a gray-scale raster image and this image is typically referred to as the *matte*. Limitations of traditional techniques include that they only key in colors in a single contiguous range of colors (for example, blues) and that they're computationally intensive.

Our technique improves on standard chroma-key techniques and allows the addition of virtual objects in the scene; for example, a virtual first-down line to a television broadcast of football, or augmenting video with virtual advertisements. The system allows for the specification of multiple colors and of colors that shouldn't be blended (that is, for foreground objects). Our approach gains efficiency by considering only those pixels that the prospective graphics effect will affect.

Pixel blending is determined by a combination of factors. The color map indicates which colors can be blended and by how much. The color map also specifies which colors shouldn't be blended. The green related to grass, for example, the white related to chalk, and the brown related to dirt can be modified, while colors in the players' uniforms should not. Figure 5 (next page) offers a basic overview of the compositing system.

The compositor considers four inputs: the program video or the foreground images; the effect or background; and an optional input α_M or control α, which can be used to modify

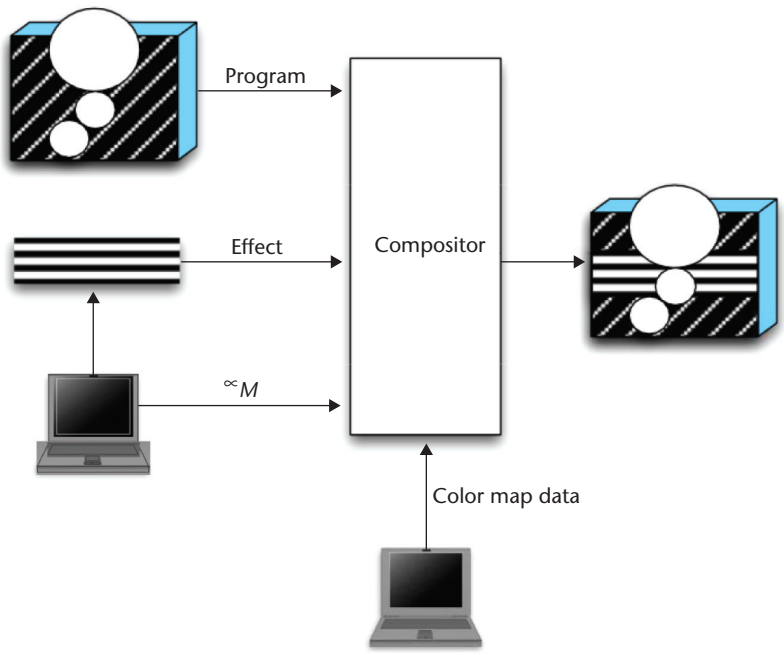


Figure 5. The compositor uses color map information to identify regions in the broadcast image where graphic effects can be added for instance on the grass of a football field and not on a player's leg.

how the effect is blended into the program video without taking into account the color in the program video. The compositor can be used to prevent blending in the video signal based on the color map. The effect typically uses designated transparent colors to provide a geometric keying effect so that the program video is only altered by nontransparent colors.

In Figure 5, the program video contains players (unmodifiable) indicated by white circles and chalk indicated by a black, diagonally striped background. The effect includes a horizontal stripe embedded in a transparent background, indicating a horizontal graphic to be added to the video image. The transparent background indicates areas in the video that are unaffected by the augmented graphics. The composite includes the horizontal line inserted behind the players.

The compositor computes the composite (C) pixels by considering the foreground pixel (FG), the background pixel (BG), α_M and α_{FG} by first computing α, and the final composite value for the composite pixel as follows:

$$\alpha = \begin{cases} \alpha_{FG} * 2 * (0.5 - \alpha_M) & \text{if } 0.0 \leq \alpha_M \leq 0.5 \\ 2 * (\alpha_M - 0.5) & \text{if } 0.5 < \alpha_M \leq 1.0 \end{cases}$$

and

$$C = (1 - \alpha) * FG + \alpha * BG \quad (1)$$

Performance can be gained by avoiding accessing the color map when $0.5 < \alpha_M \leq 1.0$. Note that when α_M is 0.5 then α becomes 0 and the background pixel (the effect) is ignored and only passes the foreground pixel to the composited image, and when α_M is 1 only the background is visible (see Equation 1). $\alpha_M = 0$ indicates that blending is determined to the value of α_{FG}. α_M is used to force either the background or the foreground pixel to be used or bias the blending.

The selection of colors to include or exclude are made using a GUI with a pointing device to indicate the blending value α_{FG} for inclusion or exclusion. The selection is then processed and inclusion and exclusion filters are generated.

Vision-based baseball tracking

We've developed techniques for tracking a baseball in two phases of play: from pitcher to plane of home base; and after a hit, from the batter to a predicted or actual impact with the ground. Here, we focus initially on how we handle the pitch phase.

Problem definition: the pitch

Our objective is to determine a parametric equation describing the motion of the ball from the pitcher to the front plane of the three-dimensional strike zone. Once we know the equation and parameters of motion, the track of the ball can be included in broadcast graphics.

An important issue therefore is the form of the equation describing the ball's motion and the relevant parameters. In his book *The Physics of Baseball*, Robert Adair describes a number of factors that affect a baseball in flight, including initial position, initial velocity, spin rate and axis, temperature, humidity, and so on.[13] In our work, we approximate Adair's equations by assuming the ball is acted upon by a constant force A throughout its flight.

The net constant force A includes the force of gravity G, the force resulting from spin S, and deceleration caused by drag (see Figure 6). Some of these forces aren't constant. For instance, the force of drag is proportional to the square of velocity, which decreases over time; and the rate of spin, which also decreases over time. Furthermore, the force arising from spin will always be perpendicular to the flight path, which is curving. And of course drag will be in line with the curving flight path.

However, we've found that assuming a constant acceleration yields good results. It's worth bearing in mind that we don't extrapolate our equations forward to find the point of impact. We use our knowledge of ball position throughout the pitch to approximate the parameters we seek.

We define the global reference frame as follows: the origin is the point of home plate nearest the catcher, $+y$ is in the direction from the plate to the pitcher, $+z$ is up, and $+x$ is to the pitcher's left. We construct equations for x, y, and z in terms of time t as follows:

$$x(t) = x_0 + v_{0x}t + \frac{1}{2}a_x t^2,$$

$$y(t) = y_0 + v_{0y}t + \frac{1}{2}a_y t^2, \text{ and}$$

$$z(t) = z_0 + v_{0z}t + \frac{1}{2}a_z t^2$$

where x_0, y_0, z_0 is the initial position of the ball at release; v_{0x}, v_{0y}, v_{0z} is the initial velocity; and a_x, a_y, a_z is the constant acceleration. Our objective then is to find the values for these nine parameters for each pitch.

Our approach is to track the ball in screen coordinates from two cameras: One above and behind home plate (high home), and one above first base (high first). Figure 7 shows the location of these cameras and provides example images captured by the cameras. Using the screen coordinates of the ball at several time instants, we're able to infer the pitch parameters.

Solving motion equations

The position, orientation, and focal length of each camera are known. Lens distortion is modeled as well. For the moment, suppose the parameters of ball motion are also known. We could then predict, at any time t, where the ball should appear in the FOV of each sensor camera. We use $P_x(t)$ to indicate the X pixel position and $P_y(t)$ to indicate the Y pixel position of the ball in the image (see Figure 7). $P_x(t)$ can be estimated in a linear equation as

$$P_x(t) = C_1x_0 + C_2y_0 + C_3z_0 + C_4v_{0x} + C_5v_{0y} + C_6v_{0z} + C_7a_x + C_8a_y + C_9a_z$$

C_1 through C_9 depend on the time and pose of the camera. However, for any particular t there are constants. C_1, C_2, and C_3 are independent of time; they're computed from the

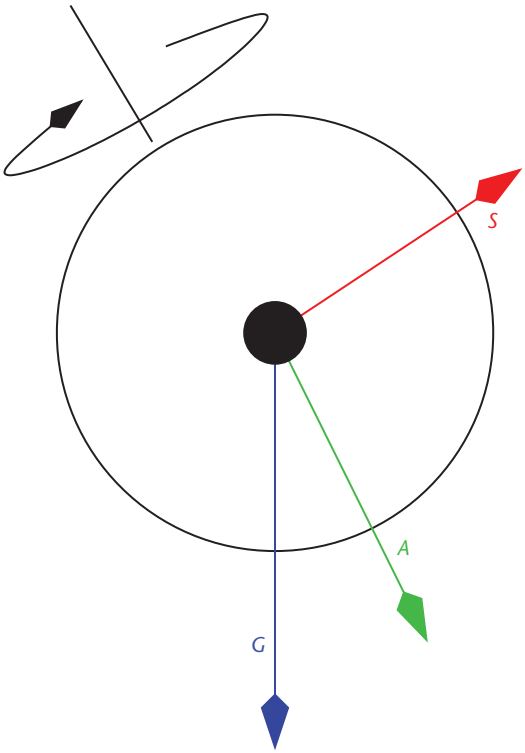


Figure 6. The forces on a baseball in flight, viewed from the pitcher's mound: G *is the force of gravity,* S *is the force resulting from the spin of the ball and its interaction with the atmosphere, and* A *is the net acceleration we solve for in our model. The deceleration caused by drag is normal to the plane of the figure.*

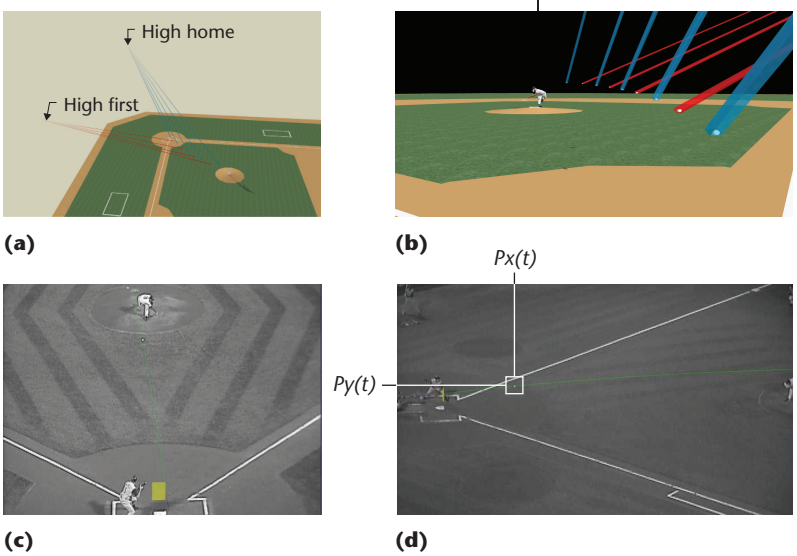
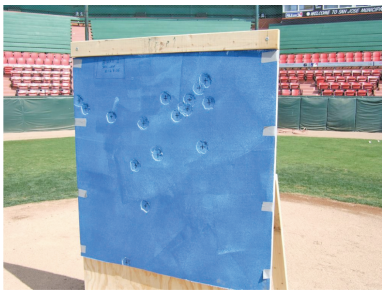


Figure 7. Using screen coordinates in two cameras to track a baseball. (a) Location of the two cameras used to track the ball during the pitch. (b) The lines of position observed by the two cameras for a pitch. (c) and (d) Example images from the high home (home plate) and high first (first base) cameras after the pitcher has released the ball, with (d) showing the X *and* Y *position of the ball at time* t *in image coordinates from the high first camera.*



(a)



(b)

Figure 8. Validating the system. (a) A pitching machine launches baseballs at (b) a foam target.

camera pose and focal length parameters and represent a projection from the global frame to the camera's frame. C_4, C_5, and C_6 depend on t and the camera pose. C_7 through C_9 depend on the camera pose and t^2. A similar equation describes the Y position of the ball from the camera's view:

$$P_y(t) = C_{10}x_0 + C_{11}y_0 + C_{12}z_0 + C_{13}v_{0x} + C_{14}v_{0y} + C_{15}v_{0z} + C_{16}a_x + C_{17}a_y + C_{18}a_z$$

The images captured at each camera are processed independently; so we have a location $P_x(t)$, $P_y(t)$ for the ball for each camera in screen coordinates at each time instance. We locate the ball in the images from each camera independently (that is, the cameras aren't synchronized).

When an image is processed to find the ball, we know t (the time the image was captured) and we find $P_x(t)$ and $P_y(t)$ by locating the ball in the image. The constants C_1 through C_{18} are computed immediately using the camera pose parameters t and t^2. Accordingly, from each image we have two equations; and each equation has 10 known values—for example, $P_x(t)$ and C_1 through C_9—and nine unknowns.

Video fields (images) are captured at 60 Hz, and the ball is in flight for about 0.5 seconds, so we collect data for 60 equations from each camera for each pitch, or 120 equations in all. The equations are solved for the nine unknowns using standard singular-value-decomposition (SVD) techniques. Because there's some ambiguity in the determination of the ball's location in each image, the SVD solution is approximated with a residual. A large residual indicates a poorly tracked pitch, in which case the data might be discarded for that pitch.

The tracking cameras

We use Bosch LTC0495 day/night cameras in black-and-white mode. The cameras include an auto iris function that we use to maintain a shutter speed of 1/1,000 or faster. The cameras include an optional IR filter. We ensure that the filter is always engaged. The cameras gather interlaced fields at 60 Hz, which our software, in turn, processes.

The software continuously searches the image for a moving ball in each camera image. We take advantage of a number of constraints to simplify the process of finding and tracking the ball. First, we limit the search in the region of each image where there are no other moving objects. The stretch of grass between the pitcher and home plate is normally devoid of players or other moving objects. Next, we use image differencing to find candidate blobs that might represent the ball. If the blob isn't round and light colored, it's discarded. Once an initial candidate blob is found, we look for a subsequent blob in a location toward home plate that corresponds with a speed of 60 to 110 miles per hour (nearly all pitches are in this range). If the track doesn't meet these requirements, it's discarded. The result is that nearly every pitch is tracked, and very few nonpitches are misidentified as pitches.

Validation, results, and use

To initially validate the system, we placed a foam board behind home plate at a local ball field. We positioned a motorized pitching machine on the pitcher's mound and launched 20 pitches at the target. Upon impact, the balls left distinct pivots in the foam board (see Figure 8). The Y location of the board was known, and we used our tracking system to predict the X and Z location of each pitch at that plane. Eighteen of the pitches hit the board (two missed and weren't considered). The pivot locations were measured carefully and compared with the predicted locations. The average error was less than 1 inch. Since that time, our system has shown noise with a standard deviation of 0.5 inch and a random bias that's diminishing with continued development to about 0.5 inch.

Figure 9 shows how the trajectory of a pitch can be reconstructed. Several example images from televised broadcasts are shown in Figure 10. The system has been employed in hundreds of MLB games for national broadcast, including the World Series in 2005. Video examples are

available at http://www.sportvision.com. Public response has been positive.

After the ball is hit

We also have a method for estimating the ball trajectory after a hit. This phase of play provides a separate challenge because any fixed camera wouldn't have a sufficient FOV to track the ball in all the locations in a stadium. In a manner similar to the techniques used for pitch tracking, we assume that the ball undergoes a constant acceleration. We further simplify the problem by assuming a fixed location (home plate) for the start of the trajectory.

We use two cameras pointed toward the home plate with fixed pan, tilt, focus, and zoom. Additional cameras are mounted on tripods having servo motors that allow computer control of the pan and tilt. Initially, the two fixed cameras capture video of the action at home plate. After the player hits the ball, we search for the ball in the video frame using image differencing. In a manner similar to that described for pitch tracking, we compute an initial velocity vector for the ball.

The ball's path can change as it travels along the initially computed trajectory due to wind, spin, and so on. Thus additional cameras are used to update the path as the ball travels. Typically we place two cameras at opposite sides of the baseball field, approximately halfway between home plate and the home run fence.

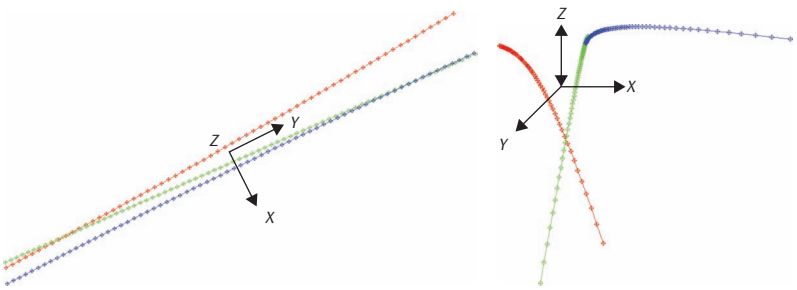


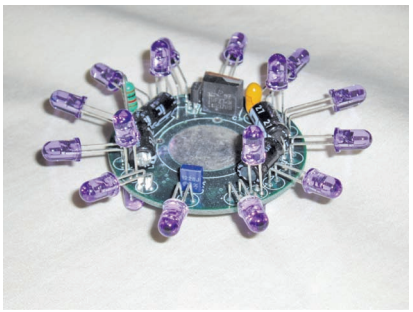
Figure 9. Trajectories of three pitches from a Major League Baseball game. For each pitch, the ball's position is indicated at 1-millisecond intervals. (a) Top-down view with the pitcher on the right and home plate on the left. (b) View of the pitches from the strike zone. Notice how the pitches seem to break dramatically at the end of their travel.

A computer uses the previously determined trajectories to predict when the ball's future path will be within the view of the outfield cameras. We predict both the time and the location along the ball's path and send signals to the servo motors of the cameras so that they point to the predicted location at the appropriate time. The outfield cameras don't move when actively capturing video.

For each outfield camera, a processor determines lines of position (as discussed previously) and the three-dimensional location of the ball at multiple times. Next, the system determines whether more data can be captured (that is, if the ball will be in view). If so, the system loops back to predict future locations and moves the



Figure 10. Several examples of the tracking system in use on broadcast television.



(a)



(b)

Figure 11. FoxTrax hockey puck. (a) Circuitry that's embedded in the pucks. (b) Several finished pucks ready for play.





Figure 12. Two images showing the on-air effect of the glowing hockey puck.

mines the location at which the ball's path would intersect the ground (field level), assuming that the ball's path isn't interrupted by a fence, a person, a wall, stadium seating, or other obstructions. This predicted position is compared to the location of home plate to determine the distance traveled, which later can be used in the broadcast.

The system provides a mechanism to display the path and the predicted path of a moving object during broadcast. For example, after the system determines the initial path of a baseball, the path can be displayed. Graphics show the location where the ball will land. As the system updates the path of the ball, the displayed path and the displayed graphics at the end of the path can be updated. For example, if the batter hits a fly ball to the outfield, while the outfielder is attempting to catch the ball the television display can show the expected path of the ball and the location in the outfield where the ball will land.

Vision-based hockey puck tracking

Although the FoxTrax hockey puck-tracking effect was our first on-air system deployed in 1996 during the NHL All-Star game, we introduce it last because it is in some ways more complex than our later tracking systems.[6] In particular, the environment in which the puck is tracked is much more challenging than is the case for baseball tracking: compare an NHL rink with the relative calm of the grass between the pitcher's mound and home plate in baseball. In addition to the players moving rapidly around, the puck is subject to direction change at any moment.

All these factors combine to make it much more difficult to find the puck and track it in video images. Our approach to this problem was to create a new puck that could be found easily. We embed bright IR LEDs inside the puck along with a circuit designed to pulse them brightly at approximately 30 Hz (see Figure 11).

Ten calibrated sensor cameras are placed in the ceiling, pointed at the ice. The sensor cameras are fitted with IR notch filters (to pass IR and block other frequencies) to enhance the signal-to-noise ratio of the flashing IR LEDs when viewed from above. All of the sensor cameras are synchronized with each other and the puck to ensure images are taken at just the right moment.

The puck's location is determined using geometry in much the same way the baseball is tracked. Given the puck's location and the known pose of each broadcast camera, we can compose a complete image with a glowing effect for the puck (see Figure 12).

The FoxTrax hockey puck was funded by Fox Sports and was first deployed (broadcast) in 1996 during the NHL All-Star game and then broadcast continuously up to the 1998 Stanley Cup Finals. Fox lost the NHL broadcast rights to ABC the following season. Fox Sports enjoyed higher viewer ratings during telecasts that included the glowing puck; this might have been because of our new technology.[6]

Future work

Our 3D-tracking techniques improve the experience for viewers watching sports ranging from baseball and hockey to NASCAR and the Olympics. Initially the FoxTrax hockey puck received mixed reviews from sports critics,[14] but follow-up technologies have been quite popular and have resulted in the award of several Emmys; for example, for NASCAR's Trackpass, football's first and ten yellow lines, and baseball's KZone. Our focus is to continue to provide the viewer with interesting information

without being intrusive. We also intend to provide interactive Web technologies for the sports viewer (other researchers are also developing such technologies[15,16]).

We're exploring other techniques for improving vision-based tracking (such as region-based mechanisms[17]) and using hybrid sensors in poor lighting conditions (such as lasers[18]) for robustness and precision. **MM**

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